Adaptive Edge Preserving Maps in Markov Random Fields for Hyperspectral Image Classification

Chao Pan, Xiuping Jia, *Senior Member, IEEE*, Jie Li, and Xinbo Gao, *Senior Member, IEEE*

***Abstract*— This article presents a novel adaptive edge preserving (aEP) scheme in Markov random fields (MRFs) for hyperspectral image (HSI) classification. MRF regularization usually suffered from over-smoothing at boundaries and insufficient refinement within class objects. This work divides and conquers this problem class-by-class, and integrates $K(K-1)/2$ (K is the class number) aEP maps (aEPMs) in MRF model. Spatial label dependence measure (SLDM) is designed to estimate the interpixel label dependence for given spectral similarity measure. For each class pair, aEPM is optimized by maximizing the difference between intraclass and interclass SLDM. Then, aEPMs are integrated with multilevel logistic (MLL) model to regularize the raw pixelwise labeling obtained by spectral and spectral–spatial methods, respectively. The graph-cuts-based $\alpha\beta$-swap algorithm is modified to optimize the designed energy function. Moreover, to evaluate the final refined results at edges and small details thoroughly, segmentation evaluation metrics are introduced. Experiments conducted on real HSI data denote the superiority of aEPMs in evaluation metrics and region consistency, especially in detail preservation.**

***Index Terms*— Edge preserving (EP), graph-cuts, hyperspectral image (HSI) classification, Markov random fields (MRFs).**

I. Introduction

REMOTE sensing offers practicable characterization of land cover in large scenes. Imaging spectroscopy [1], originated from the work by Goetz *et al.* [1] at NASA's JPL Laboratory, Pasadena, CA, USA, in the 1980s. Hyperspectral images (HSIs) have hundreds of narrow spectral bands and provide detailed electromagnetic attributes of the land-cover objects/materials.

The goal of HSI classification is to label each pixel within an observed HSI as predefined land-cover categories [2]. The land-cover mapping methods for low-dimensional data often perform poor on HSI data, which are known as the Hughes phenomena [3]. Feature extraction/selection strategies reducing input dimensionality are time-consuming and, sometimes, need a prior knowledge. In practical, kernel methods, including support vector machines (SVMs) [4], and multinomial logistic regression (MLR) [5], [6] handle the curse of dimensionality and show good generalization ability.

However, the labeling maps by spectral-based methods always appear noises. These noncontextual approaches treat HSI as a list of radiometric responses and neglect the spatial behavior pattern. The first try fusing spectral–spatial information was extraction and classification of homogenous objects (ECHO) by Kettig *et al.* [7]. Some methods, i.e., texture extraction captures the spatial repetitive patterns of local/nearby pixel intensity [8]. The morphological profile (MP) extracts contextual feature at object level [10]. Fauvel *et al.* [11] fused extended MP (EMP) and original spectral feature with SVM for accurate urban areas labeling. Moser and Serpico [12] regarded this one SVM-based spectral–spatial scheme feeding contextual features into SVM. They reviewed the other strategy, combination of SVM-based class density estimate and Markov random fields (MRFs) to apply image segmentation [12], [13].

MRFs are probabilistic graphical models in computer vision, e.g., image segmentation [14], pose tracking [15], and face recognition [16]. The Markov–Gibbs equivalence relates the Markovianity property with a Gibbs distribution [17]–[19]. The Markovianity depicts the local characteristics and in MRF, only neighboring labels have direct interactions with each other [18]. The maximum a posterior can be solved by optimizing proper energy minimization function [18]. The basic MRF model consists of two terms: the data term given by pixelwise class estimate, and the smoothness one favoring label smoothness by punishing nearby pixels of different labels [13], [18]. The label smoothness property means that the neighboring pixels have high possibility for sharing the same class label. MRFs are well-known postprocessing frameworks in remote sensing image classification [20], [21]. Recently, the basic isotropic multilevel logistic (MLL) [18] has been widely used in image classification. The MLL prior was integrated with MLR [6], and subspace MLR (MLRsub) [22] to favor label smoothness. The pixelwise labeling by multitask joint sparse representation was refined by MLL considering neighbor distance [23].

Moser *et al.* [13] pointed out the flexibility of MRF in fusing multisource/multiple information for land-cover labeling, e.g., clustering [24], and multiscale segmentation [25]. Moser and Serpico [12] unified MLL and SVM, and designed

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"Markovian kernel." In [26], MLL prior was used to further refine the raw result fusing local and global probabilities by SVM and MLRsub (SVMMLRsubMRF). The work [27] designed an anisotropic MLL fusing class co-occurrence dependence and spectral dissimilarity of pixels to promote MRF correction. In [28], spectral angle mapping (SAM) of adjacent pixels was integrated with adaptive interclass-pair penalty to provide suitable degrees of regularization. Zhang *et al.* [29] fused multiple features and nonlocal information in MRF, and modeled the spatial smoothness prior in semantic space.

The MLL prior favors area uniformity inside object, but has a symbiotic issue, preservation of details and edges [13], [18]. Previous studies for edge preserving (EP) in MRF image labeling mainly include the following types of EP maps (EPMs): accurate, binary, fuzzy, hierarchical, multiscale, semantic, and tree-structured (TS). Solberg *et al.* [30] included accurate borders by geographic information systems to improve spatial–temporal MRF classification. The accurate borders, however, are always unavailable, and the alternate EPM is based on edge detector, spectral similarity measure (SSM), or segmentation. Moser and Serpico [24] integrated a binary "edge/no-edge" map obtained by line process into MLL for EP of neighboring pixels from different clusters. Ghamisi *et al.* [31] combined a Sobel operator-based binary EPM in hidden MRF to protect discontinuities. Actually, clear borders/edges, however, often do not exist in satellite images [32] and, moreover, the binary EPMs provide too hard constraints and may be undesirable. Tarabalka *et al.* [33] fused a "fuzzy no-edge/edge" function with MLL for EP in MRF. A relative homogeneous index [34] was defined to avoid over smoothing. Spatial attraction model [35] combined the class posterior probability and distance of nearby pixels to reduce over-smoothing. The work [36] integrated Mahalanobis distance boundary with SVM in MRF. Spectral information divergence (SID) was computed on 3-D discrete wavelet transform (DWT)-based HSI feature and integrated into MRF for EP (DWTMRF) [37]. In [38], the fine structures by computing gradient were preserved in total variation regularizer. In [39], graduated increased edge penalty-based EP was designed and optimized by image region growing using semantic. In [40], "fuzzy no-border/border" weights by hierarchical statistical region segmentation tuned the MLL prior to avoid detail loss. TS-MRF [41], [42] used binary hierarchical segmentation-based edge prior to describe the hidden structure of remote sensing data and refine the raw image labeling.

The EP strategy has been a key issue in MRF label mapping. The current schemes, however, are still not (completely) class-adaptive. Even the TS-MRF has its defect that the class-based hierarchy uses binary tree and groups regions of different classes and statistics temporarily [41], [42]. Besides, the constructions (or definitions) of EPM in these methods have parameters to preset or experimentally adjust instead of preestimating the performance of expected EPM.

Note that HSI labeling suffers from low interclass and high intraclass spectral variation [43], [44], as well as these EP schemes. The former makes the detection of class borders difficult, and the latter makes EPM contain false edges inside class objects. Moreover, different neighboring classes are with various interclass/intraclass variations [43], [44].

These problems motivated us to divide and conquer the EP strategy class-by-class first for feasible analysis. Inspired by the studies on class separability [45], [46] and SSM effectiveness [47], [48], spatial label dependence measure (SLDM) is proposed and estimated with training sets. Given an SSM, the intraclass and interclass SLDM describes the label dependence of neighboring pixels within objects and at class borders. For each class pair, adaptive EPM (aEPM) is optimized by maximizing the difference between interclass and interclass SLDM for a series of SSMs. All $K(K-1)/2$ aEPMs (aEPMs are the plural forms of aEPM) customized for different class pairs are integrated with MLL in MRF model. For an overall evaluate, aEPMs (aEPMs also indicate the proposed method integrating $K(K-1)/2$ aEPMs) are used to regularize the result by spectral classifier, i.e., noncontextual SVM [4], and spectral-spatial methods, i.e., composite kernels (SVMck) [49] fusing spectral features and extended multiattribute profiles (EMAPs) [50] on HSI, respectively. The graph-cuts-based swap algorithm [51] is improved to solve the proposed energy function. Segmentation evaluation metrics [54], [55] are used for an overall assessment. The experimental results indicate the advantages of aEPMs in evaluate metrics, label uniformity, and detail preservation compared with classical MRF and spectral–spatial methods.

Though the aEPMs scheme falls into adaptive-MRF category, it is characterized by integrating of $K(K-1)/2$ aEPMs, and each aEPM is optimized by SLDM. In addition, aEPMs and classical approaches (e.g., multiscale and hierarchical) are mutually compatible and can be combined together if suitable. This work mainly includes the following innovations.

1) This article designs SLDM to evaluate the effectiveness of a given SSM. For one class pair, SLDM is optimized by maximizing the difference between intraclass SLDM and interclass SLDM given a series of SSMs, and then the customized aEPM gets a tradeoff between favoring label uniformity inside objects and protecting class boundaries.
2) The study proposes a class-adaptive aEP scheme, which divides and conquers EP class-by-class, and combines $K(K-1)/2$ aEPMs in MRF model to avoid over-smoothing at details and favor label smoothness within objects.
3) This work extends the $\alpha\beta$-swap algorithm to solve the proposed energy function with $K(K-1)/2$ aEPMs.
4) Several segmentation evaluation metrics are introduced, which evaluate the effectiveness of comparative methods on preserving small details and boundary regions.

This article is organized as follows. Section II presents the formulation of MRF-based HSI classification. The key ideas on the proposed aEPMs scheme are shown in Section III. Section IV presents the experimental results of comparative methods. Section V draws the conclusions and discussions.

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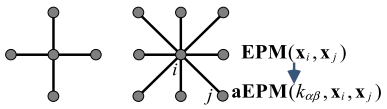


Fig. 1. First- and second-order neighborhood systems.

II. Problem Formulation

The goal of classification is to interpret the acquired HSI $\mathbf{x}$ as a mapping $\mathbf{y}$, where $\mathbf{y}$ means the land-cover classes. We denote one pixel coordinate in an n-pixel image as $i \in \{1, 2, \ldots, n\}$. The spectral attribute and the label of the ith pixel are $\mathbf{x}_i$ and $y_i \in \mathbf{K} = \{1, 2, \ldots, K\}$, respectively, where K is the number of classes. MRF has been widely applied in HSI classification [13]. The Hammersley–Clifford theorem [18] establishes the equivalence between the local property of an MRF and the global property (Gibbs distribution) of a Gibbs random field

$$p(\mathbf{y}|\mathbf{x}) = \frac{1}{Z}\exp\{-\mathbf{U}(\mathbf{y}|\mathbf{x})\} \tag{1}$$

where $\mathbf{U}(\mathbf{y}|\mathbf{x})$ is known as potential energy function and defined with neighboring system, as shown in Fig. 1. And Z is the partition function [18]. Then, the maximum a posterior estimate of $p(\mathbf{y}|\mathbf{x})$ is equivalent to minimizing $\mathbf{U}(\mathbf{y}|\mathbf{x})$. A basic MRF model contains two terms

$$\mathbf{U}(\mathbf{y}|\mathbf{x}) = \mathbf{U}_{\text{data}} + \mathbf{U}_{\text{smooth}} \tag{2}$$

$$\mathbf{U}_{\text{data}} = -\sum_{i=1}^{n} \log p(y_i|\mathbf{x}_i) \tag{3}$$

$$\mathbf{U}_{\text{smooth}} = \mu \sum_{j\in N_i} \left[1 - \delta\left(y_i, y_j\right)\right] \tag{4}$$

where $\mathbf{U}_{\text{data}}$ denotes class estimate by pixelwise probabilistic classification. $\delta(y_i, y_j)$ is the impulse function, and $\delta(y_i, y_j) = 1$ for $y_i = y_j$, and $\delta(y_i, y_j) = 0$ for $y_i \neq y_j$. $\mathbf{U}_{\text{smooth}}$ indicates smoothness term favoring label smoothness. The value μ tunes the mutual importance of spectral and spatial terms.

III. Proposed Framework

The flowchart of aEPMs is shown in Fig. 2. Totally, $K(K-1)/2$ aEPMs are optimized by SLDM. All aEPMs are integrated with MLL during MRF correction. The class probability is estimated via probabilistic classification. A modified $\alpha\beta$-swap algorithm is used to optimize the designed energy function.

A. Class Probability Estimate

SVM has been widely used nonparametric statistic method in land-cover mapping [4]. Probabilistic SVM [56] is used as the basic classifier in this work to estimate $\mathbf{U}_{\text{data}}$ in (3). The typical nonlinear SVM is defined as follows [4]:

$$f(\mathbf{x}_{\text{test}}) = \sum_{\mathbf{x}_i \in \mathbf{S}_{\text{train}}} \ell_i y_i \mathbf{K}(\mathbf{x}_i, \mathbf{x}_{\text{test}}) + b \tag{5}$$

where $\mathbf{S}_{\text{train}}$ denotes the training set and b is bias. $\mathbf{x}_{\text{test}}$ is a test sample. The Lagrange multipliers $\{\ell_i\}$ weight the importance of training samples in discrimination function. Kernel methods [57] validate SVM superior performance in HSI classification.

The spatial features are critical for discriminating structured and textured categories, e.g., roads, buildings, and agriculture scenes [11]. Moser *et al.* [13] shown the use of textured features improved the labeling by raw classifier, and then benefited the following MRF correction. Moser *et al.* [13] have suggested the integration of MRF, mathematical morphology, and SVM.

In this work, EMAP [50] is utilized to extract spatial feature. Morphological attribute profile (AP) [58] was generalized from MP [10]. Extended attribute profiles (EAPs) [50] are defined as a list of APs computed on the c principle components (PCs) extracted from the HSI data

$$\text{EAP} = \{\text{AP}(\text{PC}_1), \text{AP}(\text{PC}_2), \ldots, \text{AP}(\text{PC}_c)\}. \tag{6}$$

EAP has facility over EMP by defining different attributes to model the needed spatial features. An EMAP [50] containing m different EAPs is given as the following format:

$$\text{EMAP} = \left\{\text{EAP}_{a_1}, \text{EAP}_{a_2}, \ldots, \text{EAP}_{a_m}\right\} \tag{7}$$

where $\{a_1, a_2, \ldots, a_m\}$ denotes the m different attributes. EMAP is powerful in extracting geospatial features compared with EAP.

To estimate$\mathbf{U}_{\text{data}}$, SVMck integrates the spectral and EMAP features using weighted summation kernel [49] as follows:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \lambda K_{\text{spec}}\left(\mathbf{x}_i^{\text{spec}}, \mathbf{x}_j^{\text{spec}}\right) + (1-\lambda) K_{\text{EMAP}}\left(\mathbf{x}_i^{\text{EMAP}}, \mathbf{x}_j^{\text{EMAP}}\right) \tag{8}$$

where $\mathbf{x}_i^{\text{spec}}$ and $\mathbf{x}_i^{\text{EMAP}}$ denote the spectral and EMAP features of the ith pixel, and K_{spec} and K_{EMAP} are the corresponding kernel functions, and λ tunes the spectral and spatial content.

For better data separability, a Gaussian radial basis function (RBF) $K_{\text{spec/EMAP}}(\mathbf{x}_i, \mathbf{x}_j) = \exp\{-\gamma ||\mathbf{x}_i - \mathbf{x}_j||^2\}$ is used and γ tunes the width of the Gaussian kernel. Both SVM applied on spectral features, and SVMck with spectral and EMAP feature are utilized to estimate $\mathbf{U}_{\text{data}}$, respectively.

B. Spatial Label Dependence Measure

SLDM is one main innovation in this article. A pointcut is selected from existing EP methods. First, the hierarchical or multiscale EP schemes can be seen as an integration of binary or fuzzy EPMs, because for each hierarchy or one scale, EPM is either binary or fuzzy. Second, accurate edges are always unavailable in HSI data, particularly for agriculture scenes, vegetation, and soil in urban area, and the binary EPM can be regarded as thresholding the fuzzy one. Third, various methods were used to exploit the interpixel dependence to construct fuzzy EPM. Finally, we recall that SSM has been widely used in HSI analysis [47], [48], including EP strategies in MRF.

In terms of the above analysis, the basic SSM-based EP in MRF regularization is selected as a pointcut, as given by

$$\mathbf{U}_{\text{smooth}}^{\text{EPM}} = \mu \sum_{j\in N_i} \mathbf{EPM}\left(\mathbf{x}_i, \mathbf{x}_j\right)\left[1 - \delta\left(y_i, y_j\right)\right] \tag{9}$$

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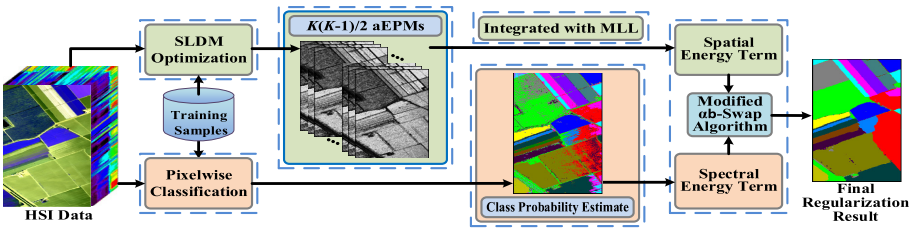


Fig. 2. Flow diagram of the proposed aEP integrating $K(K-1)/2$ aEPMs, each of which is optimized by SLDM.

where $\mathbf{EPM}(\mathbf{x}_i, \mathbf{x}_j)$ denotes SSM of pixels i and j as follows:

$$\mathbf{EPM}(\mathbf{x}_i, \mathbf{x}_j) = \exp\{-d^2(\mathbf{x}_i, \mathbf{x}_j)\} \tag{10}$$

$$d(\mathbf{x}_i, \mathbf{x}_j) = \mathrm{SAM}(\mathbf{x}_i, \mathbf{x}_j) = \cos^{-1}\frac{<\mathbf{x}_i, \mathbf{x}_j>}{||\mathbf{x}_i||_2||\mathbf{x}_j||_2} \tag{11}$$

where SAM is selected as SSM for coping the variations in incoming for radiation energy, such as in the shaded areas. For pixels with similar spectral feature, EPM makes the interpixel penalty larger to favor label smoothness and vice versa.

Consider a binary land-cover mapping task with classes α and β, and we modified (10) as follows:

$$\mathbf{aEPM}(k_{\alpha\beta}, \mathbf{x}_i, \mathbf{x}_j) = \exp\{-d^2(\mathbf{x}_i, \mathbf{x}_j)/k_{\alpha\beta}\} \tag{12}$$

where $k_{\alpha\beta} > 0$ tunes the sensitivity of aEPM to the change of SSM, which has been used in [36] and [39]. As $k_{\alpha\beta}$ increases, the false edges within classes tend to be refined sufficiently and, however, over-smoothing risk of true class boundaries between different land covers also increases.

MRF correction expects refinement of misclassified isolated pixels within class objects and, meanwhile, protects edges and details. Given a series of $k_{\alpha\beta}$, the possibility of detecting false edges in class regions and true edges at boundary areas is associated with the intraclass and interclass variabilities.

For a set of training samples of classes α and class β, $\mathbf{x}_\alpha \equiv \{\mathbf{x}_1, \ldots, \mathbf{x}_{n_\alpha}\}$ and $\mathbf{x}_\beta \equiv \{\mathbf{x}_1, \ldots, \mathbf{x}_{n_\beta}\}$, where n_α and n_β represent the numbers of training samples for classes α and β, and the interclass SLDM is given as

$$\mathbf{SLDM}^{\mathrm{Inter}}_{\alpha\beta}(k_{\alpha\beta}) = \frac{1}{n_\alpha n_\beta}\sum_{\mathbf{x}_i\in\mathbf{X}_\alpha, \mathbf{x}_j\in\mathbf{X}_\beta}\mathbf{aEPM}(k_{\alpha\beta}, \mathbf{x}_i, \mathbf{x}_j) \tag{13}$$

where $\mathbf{SLDM}^{\mathrm{Inter}}_{\alpha\beta}$ is the interpixel dependence at borders given $k_{\alpha\beta}$. See that it is estimated by applying (13) on the training set and is an approximation of interclass variability.

Correspondingly, the intraclass SLDM given $k_{\alpha\beta}$ for class α is predicted by approximating intraclass spectrum variability

$$\mathbf{SLDM}^{\mathrm{Intra}}_{\alpha\alpha}(k_{\alpha\beta}) = \frac{1}{n_\alpha(n_\alpha-1)/2}\sum_{\{\mathbf{x}_i,\mathbf{x}_j\}\subseteq\mathbf{X}_\alpha}\mathbf{aEPM}(k_{\alpha\beta}, \mathbf{x}_i, \mathbf{x}_j) \tag{14}$$

where $\mathbf{SLDM}_{\alpha\alpha}$ means the interpixel spatial label dependence within homogeneous class objects given $k_{\alpha\beta}$.

Both $\mathbf{SLDM}_{\alpha\beta}$ and $\mathbf{SLDM}_{\alpha\alpha}$ increase when $k_{\alpha\beta}$ decreases. Fortunately, the studies on class spectral separability measure [45], [46] and SSM effectiveness [47], [48] inspired us. Class separability measure has been used for feature selection, such as scatter-matrix-based measure [46] and Fisher ratio [45]. A good class separability measured by scatter-matrix expects small within-class scattering but large between-class scattering [46]. And Fisher ratio is obtained as the ratio of the interclass distance to the intraclass difference [45].

Then, analyze the evaluation of SSM criterion, e.g., spectral discriminatory probability and spectral discriminatory power [47], [48]. Good discriminatory performance of SSM expects small intraclass distance but large interclass distance.

As stated, (13) and (14) are defined by approximating the intraclass and interclass variation, and to assess the possibility of detecting false edges with class objects and true boundaries between different land covers. Similarly, **SLDM** for α and β is defined as the difference between (14) and (13)

$$\begin{aligned}\mathbf{SLDM}(k_{\alpha\beta}) = &\left[\mathbf{SLDM}^{\mathrm{Intra}}_{\alpha\alpha}(k_{\alpha\beta}) - \mathbf{SLDM}^{\mathrm{Inter}}_{\alpha\beta}(k_{\alpha\beta})\right]\\ &+ \left[\mathbf{SLDM}^{\mathrm{Intra}}_{\beta\beta}(k_{\alpha\beta}) - \mathbf{SLDM}^{\mathrm{Inter}}_{\beta\alpha}(k_{\alpha\beta})\right]\end{aligned} \tag{15}$$

where $\mathbf{SLDM}(k_{\alpha\beta}) \in (0, 2)$ describes the goodness of tradeoff between favoring label uniformity and protecting boundaries. Large values of $\mathbf{SLDM}(k_{\alpha\beta})$ expect a good tradeoff of label smoothness and detail preservation in MRF regularization.

C. aEP Maps

Consider only one aEPM which is optimized by maximizing the summation of SLDMs for all combinations of classes

$$k^{\mathrm{opt}} = \arg\max_{k_{\alpha\beta}}\sum_{\{\alpha,\beta\}\subseteq\mathbf{K}}\mathbf{SLDM}(k_{\alpha\beta}) \tag{16}$$

$$\mathbf{U}^{\mathrm{aEPM^S}}_{\mathrm{smooth}} = \mu\sum_{j\in N_i}\mathbf{aEPM}(k^{\mathrm{opt}}, \mathbf{x}_i, \mathbf{x}_j)\left[1-\delta(y_i, y_j)\right] \tag{17}$$

where $\mathbf{U}^{\mathrm{aEPM^S}}_{\mathrm{smooth}}$ is the improved smoothing term compared with (4).

This case is a simplification of aEPMs, the so-called aEPM^{S}. However, aEPM^{S} is a tradeoff for all possible adjacent classes. It is obvious that SLDM can be customized class-by-class.

$$k^{\mathrm{opt}}_{\alpha\beta} = \arg\max_{k_{\alpha\beta}}\mathbf{SLDM}(k_{\alpha\beta}). \tag{18}$$

Then, all $K(K-1)/2$ aEPMs are integrated into MLL and the designed smoothness term is given as follows:

$$\begin{aligned}&\mathbf{U}^{aEPM_{\alpha\beta}}_{\mathrm{smooth}}\\ &= \mu\sum_{\substack{y_i, y_j\in\{\alpha,\beta\}\\ j\in N_i}}\mathbf{aEPM}\left(k^{\mathrm{opt}}_{\alpha\beta}, \mathbf{x}_i, \mathbf{x}_j\right)\left[1-\delta(y_i, y_j)\right]\end{aligned} \tag{19}$$

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$$\mathbf{U}_{\text{smooth}}^{\text{aEPMs}} = \mu \sum_{\substack{\{\alpha,\beta\}\subseteq K \\ y_i, y_j \in \{\alpha,\beta\} \\ j \in N_i}} \mathbf{aEPM}\left(k_{\alpha\beta}^{\text{opt}}, \mathbf{x}_i, \mathbf{x}_j\right)\left[1-\delta\left(y_i, y_j\right)\right] \tag{20}$$

where $\mathbf{U}_{\text{smooth}}^{\text{aEPM}_{\alpha\beta}}$ is the MRF smoothness term related to α and β, and $\{\alpha, \beta\} \subseteq \mathbf{K}$ in (20) generalizes $\mathbf{U}_{\text{smooth}}^{\text{aEPM}_{\alpha\beta}}$ to multiclass task.

Combine (3) with (17) and (20), respectively, and the novel energy functions for aEPM^{S} and aEPMs are given as follows:

$$\begin{aligned}\mathbf{U}^{\text{aEPM}^{\text{S}}}(\mathbf{y}|\mathbf{x}) &= \mathbf{U}_{\text{data}} + \mathbf{U}_{\text{smooth}}^{\text{aEPM}^{\text{S}}} \\ &= -\sum_{i=1}^{n} \log p(y_i|\mathbf{x}_i) \\ &\quad + \mu \sum_{j\in N_i} \mathbf{aEPM}\left(k^{\text{opt}}, \mathbf{x}_i, \mathbf{x}_j\right)\left[1-\delta\left(y_i, y_j\right)\right]\end{aligned} \tag{21}$$

$$\begin{aligned}\mathbf{U}^{\text{aEPMs}}(\mathbf{y}|\mathbf{x}) &= \mathbf{U}_{\text{data}} + \mathbf{U}_{\text{smooth}}^{\text{aEPMs}} \\ &= -\sum_{i=1}^{n} \log p(y_i|\mathbf{x}_i) \\ &\quad + \mu \sum_{\substack{\{\alpha,\beta\}\subseteq K \\ y_i, y_j \in \{\alpha,\beta\} \\ j \in N_i}} \mathbf{aEPM}\left(k_{\alpha\beta}^{\text{opt}}, \mathbf{x}_i, \mathbf{x}_j\right)\left[1-\delta\left(y_i, y_j\right)\right].\end{aligned} \tag{22}$$

The proposed aEPMs integrate $K(K-1)/2$ aEPMs with MLL considering all combinations of class pairs $\{\alpha, \beta\} \subseteq \mathbf{K}$, and each aEPM for the class pair $\{\alpha, \beta\}$ is optimized by maximizing SLDM. Moreover, SLDM is associated with the intraclass/interclass spectrum variability. Hence, the optimal aEPM for each class pair is adopted during MRF regularization to offer enough smoothness within homogeneous class areas and, meanwhile, avoid over-smoothing at class edges and details. This makes the aEPMs superior to traditional EP schemes [33]–[40], where the parameters of EPM need to be preset or experimentally tuned. Moreover, compared with the TS-MRF [41], [42] using binary tree and grouping regions of different classes temporarily, aEPMs divide and conquer the EP problem class-by-class as the EP parameter $k_{\alpha\beta}$ is defined in (12) and optimized by (18). In addition, for aEPM^{S}, one aEPM is computed by maximizing the summation of SLDMs over all $\{\alpha, \beta\} \subseteq \mathbf{K}$.

D. Optimization Algorithm

There are many methods for energy minimization in MRF image labeling, including iterated conditional mode [60], and simulated annealing [61]. Recently, graph-cuts-based methods [51] have attracted lots of interest in HSI classification for fast approximation and ideal solutions [6], [9], [22], [26], [28], [37]. Here, graph-cuts-based $\alpha\beta$-swap method [51] is introduced and modified for energy optimization. Within one $\alpha\beta$-swap, for classes $\{\alpha, \beta\} \subset \mathrm{K}$, the $\alpha\beta$-swap move swaps the labels between an arbitrary set of pixels labeled α and another arbitrary set that labeled β [51]. See that for any two classes $\{\alpha, \beta\} \subseteq \mathbf{K}$, it holds that $V(\alpha, \alpha) + V(\beta, \beta) \leq V(\alpha, \beta) + V(\beta, \alpha)$ [52].

The optimization of aEPMs is described in Algorithm 1. The data term $\mathbf{U}_{\text{data}}$ is given by raw probabilistic classification, as illustrated detailly in Section III-A. The smoothness term $\mathbf{U}_{\text{smooth}}^{\text{aEPMs}}$ is integrated with $K(K-1)/2$ aEPMs, each of which is optimized class-by-class using SLDM, whereas the original $\alpha\beta$-swap algorithm deals with only one EPM from beginning to end. The optimization begins with randomly selected labeling $\mathbf{y}^{\text{start}}$ and operates class-pair-by-class-pair within one iteration.

Algorithm 1 Modified $\alpha\beta$-Swap Algorithm for aEPMs

Initialization:
Spectral energy term: $\mathbf{U}_{\text{data}}$;
Spatial energy term: $\mathbf{U}_{\text{smooth}}^{\text{aEPMs}}$ integrated $K(K-1)/2$ aEPMs;
Parameter weighting spectral and spatial energy: μ;
Number of the total $\alpha\beta$-swap iterations: N^{total};
Index of the current $\alpha\beta$-swap iteration: N^{index};
1. Start with an arbitrary labeling $\mathbf{y} := \mathbf{y}^{\text{start}}$ and $N^{\text{index}} := 0$;
2. **Recycle**
3. $N^{\text{index}} := N^{\text{index}} + 1$;
4. **Set** Mark: = False;
5. **For** one class pair $\{\alpha, \beta\} \subseteq \mathbf{K}$.
   5.1 **Set** the current aEPM for $\{\alpha, \beta\} \subseteq \mathbf{K}$;
   5.2 **Update** $\mathbf{U}^{\text{aEPMs}}(\mathbf{y}|\mathbf{x})$ on the current aEPM;
   5.3 **Find** $\hat{\mathbf{y}} = \arg\min \mathbf{U}^{\text{aEPMs}}(\mathbf{y}'|\mathbf{x})$ among $\mathbf{y}'$ within one $\alpha\beta$-swap operation over$\mathbf{y}$;
   5.4 **If** $\mathbf{U}^{\text{aEPMs}}(\mathbf{y}'|\mathbf{x}) < \mathbf{U}^{\text{aEPMs}}(\mathbf{y}|\mathbf{x})$, **Set** $\mathbf{y} := \mathbf{y}'$ and Mark := True;
6. **If** Mark = True **Goto** 4;
7. **Until** $N^{\text{index}} = N^{\text{total}}$;
8. **Return** $\mathbf{y}$;

See that during one cycle when switching from another class pair to the current one, i.e., $\{\alpha, \beta\} \subseteq \mathbf{K}$, the current aEPM is converted to the one customized for $\{\alpha, \beta\}$, and the value of $\mathbf{U}^{\text{aEPMs}}(\mathbf{y}|\mathbf{x})$ is updated according to the newly selected aEPM. This improvement enables the modified $\alpha\beta$-swap method to handle multiple aEPMs. The algorithm will stop after reaching the cycle number N^{total}. In addition, for aEPM^{S}, one aEPM given by (17) is used throughout the optimization process.

IV. Experimental Results

In this article, three real HSI data shown in Fig. 3, the AVIRIS Salinas Scene, the ROSIS Pavia University, and a part of the HSI data used in the 2018 IEEE GRSS Data Fusion Contest (grss_dfc_2018) [62], [63] are adopted for comparative experiments. The Salinas Scene covered Salinas Valley and has 204 bands after removing water absorption bands. This scene contains 16 classes and 512 × 217 pixels. The Pavia University has 610 × 340 samples and nine classes. Its resolution is 1.3 m. From the grss_dfc_2018 data, a subregion

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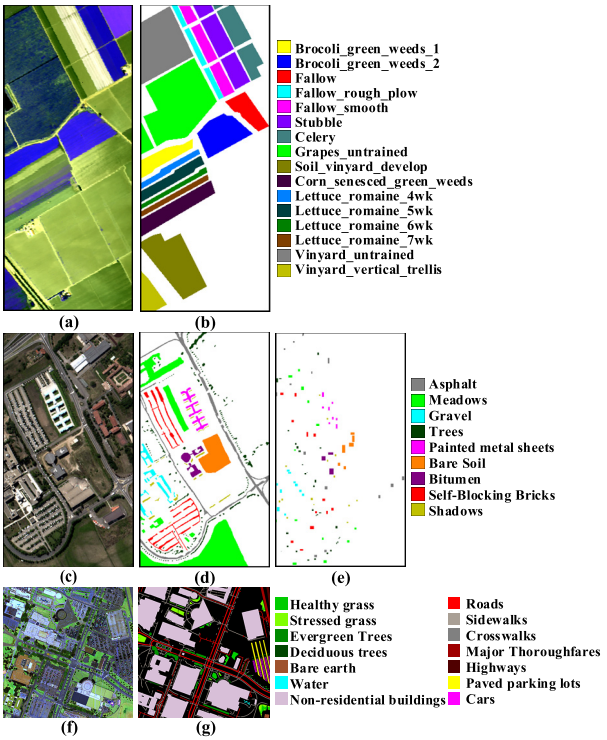


Fig. 3. (a), (c), and (f) False color images, (b), (d), and (g) ground truth for Salinas Scene, Pavia University, and DFC2018sub, and (e) FTS for (d).

(DFC2018sub) was selected by moving a 601 × 601 pixels window on the ground truth regions, where there are the most labeled pixels. It covers 380–1050-nm spectral range with 48 bands. Its resolution is 1 m and it contains 14 classes.

The proposed aEPM^S and aEPMs, and classical methods, MLL, EPM given by (11), are used to refine the raw results by SVM and SVMck, respectively. One well-known probabilistic SVM [56] in LibSVM [59] is used to offer class probability estimate. The compared MRF methods include MLRsubMLL [22], SVMMLRsubMRF [26], and DWTMRF [37]. The other spectral–spatial methods, generalized composite kernel (MLR-GCK) [64], discontinuity preserving relaxation (DPR) [65], and principal component analysis (PCA)-based edge-preserving features (PCAEPFs) [66] are used for comparison and the parameters are as in [64]–[66].

A. Evaluation Metrics

The evaluation metrics are individual class accuracy (ICA, also known as recall), overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa).

However, these metrics are conducted on limited available labeled samples inside homogeneous areas, and insufficient to evaluate MRF regularization at boundaries and small details. Thus, this article introduces segmentation evaluate metrics, such as weighed variance (Var) [54], [55], and length of edge (LoE).

Denote the final result map as segmentation $\mathbf{S} = \{S_1, \ldots, S_{n_s}\}$, where S_i is a connected region of the same class. Var measures the intrasegment homogeneity

$$\mathrm{Var} = |S_i| * \mathrm{var}_i / \sum_{S_i \in \mathbf{S}} |S_i| \tag{23}$$

TABLE I

PARAMETER μ FOR SALINAS SCENE, PAVIA UNIVERSITY, AND DFC2018SUB

HSI Data	SVM				SVMck				MLRsub	SVMMLR
	MLL	EPM	aEPM^S	aEPMs	MLL	EPM	aEPM^S	aEPMs	MLL	subMRF
Salinas Scene	2.00	2.00	5.25	9.00	2.50	2.50	4.25	8.50	0.25	2.00
Pavia University	3.50	4.00	13.50	7.50	3.25	3.50	3.75	3.00	1.50	1.75
DFC2018sub	2.75	2.75	4.00	7.25	2.00	2.25	2.50	4.75	4.50	4.50

where var_i is the variance, and $|S_i|$ is pixel number in segment S_i, respectively. Considering the area of regions, Var is robust to the change of Var within small segments. Low values of Var mean overall homogeneous segmentation map [54], [55].

Moreover, LoE is defined to assess the goodness of edge and detail preserving. By gray-level co-occurrence matrix [67], LoE indicates the frequency that different labels occur adjacently

$$\mathrm{LoE} = \sum_{\{\alpha,\beta\} \subseteq \mathbf{K}} |\{(i,j) | j \in N_i, (y_i, y_j) = (\alpha, \beta), \alpha \neq \beta, i \geq j\}| \tag{24}$$

where the condition $\{\alpha, \beta\} \subseteq \mathbf{K}$ considers all possible adjacent cases of different classes. For raw pixelwise classification maps with noises, Large values of LoE indicate more fragmentation of mislabeled pixels within class regions. After appropriate postprocessing (e.g., MRF-based refinement) on raw pixelwise result, the value of LoE decreases. However, when the pixelwise result is over-regularized, the value of LoE continues to decrease because edges and small details are over-smoothness.

B. Experiments on Salinas Scene

The parameter μ and the number of selected samples are tabulated in Table I and Table II, respectively. For Salinas Scene, 1% randomly selected samples from the ground truth were used for training and the rest for test, as shown in Table II. For SVM, the parameters C tuning the amount of penalty and γ adjusting the spread of RBF kernel are set as 10^2 and 2^{-6}, respectively. For EMAP, two attributes, area and standard deviation of the regions (EMP_a and EMP_s), are used and the threshold in the AP filtering is $\lambda_a = [100\ 200\ 500\ 1000]$ and $\lambda_s = [2.5\ 5\ 7.5\ 10]$, respectively. The λ_a and λ_s values are experimentally tuned according to [50] and [58]. For SVMck, λ tuning the spectral and EMAP features in the summation kernel was set as $\lambda = 0.5$ and the values of C and γ were tuned by two cross-fold validation in $[2^{-4}, 2^{16}]$ and $[2^{-12}, 2^3]$. The threshold λ is experimentally adjusted according to [49]. The settings of λ_a, λ_s, and λ may be not the optimal parameters, under which SVMck presents ideal classification results (compared with the results in [64]) for the following MRF-based regularization. For aEPM^S and aEPMs, including the compared MRF methods, the parameter μ is experimentally tuned, as shown in Table I, when the best OA is obtained. For DWTMRF, the value of μ is set as suggested in [37]. The values of k^{opt} and $k^{\mathrm{opt}}_{\alpha\beta}$ are solved from $[2^{-40}, 2^{-40+1/4}, \ldots, 2^{10-1/4}, 2^{10}]$ by (16) and (17).

The classification maps and evaluation metrics (mean values over 100 random tests) are shown in Fig. 4 and Table II. The experimental results are analyzed detailly as follows.

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TABLE II

EVALUATION METRICS ON SALINAS SCENE OBTAINED BY DIFFERENT METHODS

Class		Train/Test	SVM					SVMck					MLRsub	SVMMLR	MLRGCK	DPR	DWTMRF	PCA
			Raw	MLL	EPM	aEPM^S	aEPMs	Raw	MLL	EPM	aEPM^S	aEPMs	MLL	subMRF				EPFs
C1	Weeds_1	20/1989	97.64	99.97	99.97	**100**	**100**	99.30	**100**	**100**	**100**	**100**	98.37	99.86	99.90	99.83	97.98	**100**
C2	Weeds_2	37/3689	99.34	99.92	99.92	**99.99**	99.82	99.68	**100**	**100**	**100**	99.87	99.15	99.78	99.39	**100**	98.73	**100**
C3	Fallow	20/1956	93.99	99.83	99.83	**100**	99.82	97.92	99.89	99.89	**99.90**	99.80	5.57	87.80	99.76	99.72	92.80	**100**
C4	**Fallow_plow**	14/1380	97.44	**97.59**	97.43	73.28	97.33	98.22	94.05	93.80	80.24	**98.72**	44.51	97.97	98.33	98.93	97.13	**99.34**
C5	**Fallow_smooth**	27/2651	97.13	98.38	98.35	99.42	**99.56**	97.77	98.65	98.67	99.51	**99.54**	97.65	99.25	98.82	99.14	99.07	98.83
C6	Stubble	40/3919	99.49	**99.98**	**99.98**	99.88	99.87	99.62	**99.95**	**99.95**	99.90	99.91	99.72	99.85	99.23	99.97	99.03	99.89
C7	Celery	36/3543	99.38	99.74	99.75	99.66	**99.76**	99.63	**99.97**	**99.97**	99.79	99.77	99.71	99.74	99.28	**100**	98.59	**100**
C8	**Grapes_untrained**	113/11158	86.14	98.98	98.98	99.26	**99.48**	89.81	97.71	97.69	97.59	**97.86**	95.20	96.74	90.96	99.17	94.66	**99.70**
C9	Soil	62/6141	99.03	99.74	99.75	**99.92**	99.68	99.51	99.96	99.97	**100**	99.80	99.41	99.69	99.87	**100**	98.93	**100**
C10	Corn	33/3245	88.54	96.07	96.05	97.50	**97.93**	92.97	97.50	97.51	97.53	**97.76**	89.45	93.65	95.91	97.85	93.90	**99.58**
C11	Lettuce_4wk	11/1057	86.39	94.47	94.47	95.03	**96.73**	90.84	96.28	96.34	96.94	**98.70**	15.50	94.75	95.60	**99.62**	93.32	99.08
C12	Lettuce_5wk	19/1908	97.44	**100**	**100**	**100**	**100**	99.58	**100**	**100**	**100**	**100**	96.13	99.96	99.95	99.47	99.31	99.86
C13	Lettuce_6wk	9/907	**96.16**	93.94	94.38	95.95	89.96	**97.82**	92.67	92.79	97.76	93.77	98.84	98.88	98.73	**99.00**	91.24	98.42
C14	Lettuce_7wk	11/1059	89.98	94.63	94.67	**96.15**	85.58	92.83	96.83	96.95	**97.33**	96.43	92.92	95.07	94.26	97.14	91.56	97.19
C15	**Vinyard_untrained**	73/7195	56.76	56.98	57.02	77.04	**89.29**	75.28	90.21	90.19	96.99	**98.23**	34.35	44.66	88.05	99.32	80.69	**99.43**
C16	Vinyard_trellis	18/1789	92.71	96.66	96.66	98.18	**98.55**	95.65	98.83	98.84	99.07	**99.36**	96.71	97.45	94.40	98.25	95.32	**99.94**
OA (Mean in %)			89.00	93.14	93.15	95.55	**97.59**	93.23	97.52	97.52	98.20	**98.84**	82.25	90.58	95.58	99.36	94.47	**99.63**
Recall/AA (Mean in %)			92.35	95.43	95.45	95.70	**97.08**	95.40	97.66	97.66	97.66	**98.72**	78.95	94.07	97.03	99.21	95.14	**99.44**
Kappa			87.71	92.32	92.33	95.02	**97.30**	92.45	97.24	97.23	98.00	**98.71**	80.02	89.45	95.08	99.29	93.83	**99.59**
Var (Mean in E+06)			5.133	8.379	8.230	**7.355**	7.619	4.193	8.340	8.223	**6.774**	7.072	6.138	6.599	4.529	7.177	8.599	11.181
LoE (Mean in E+04)			7.795	1.158	1.165	1.245	**1.476**	6.124	1.053	1.060	1.353	**1.591**	2.734	1.959	5.028	1.466	1.772	1.411

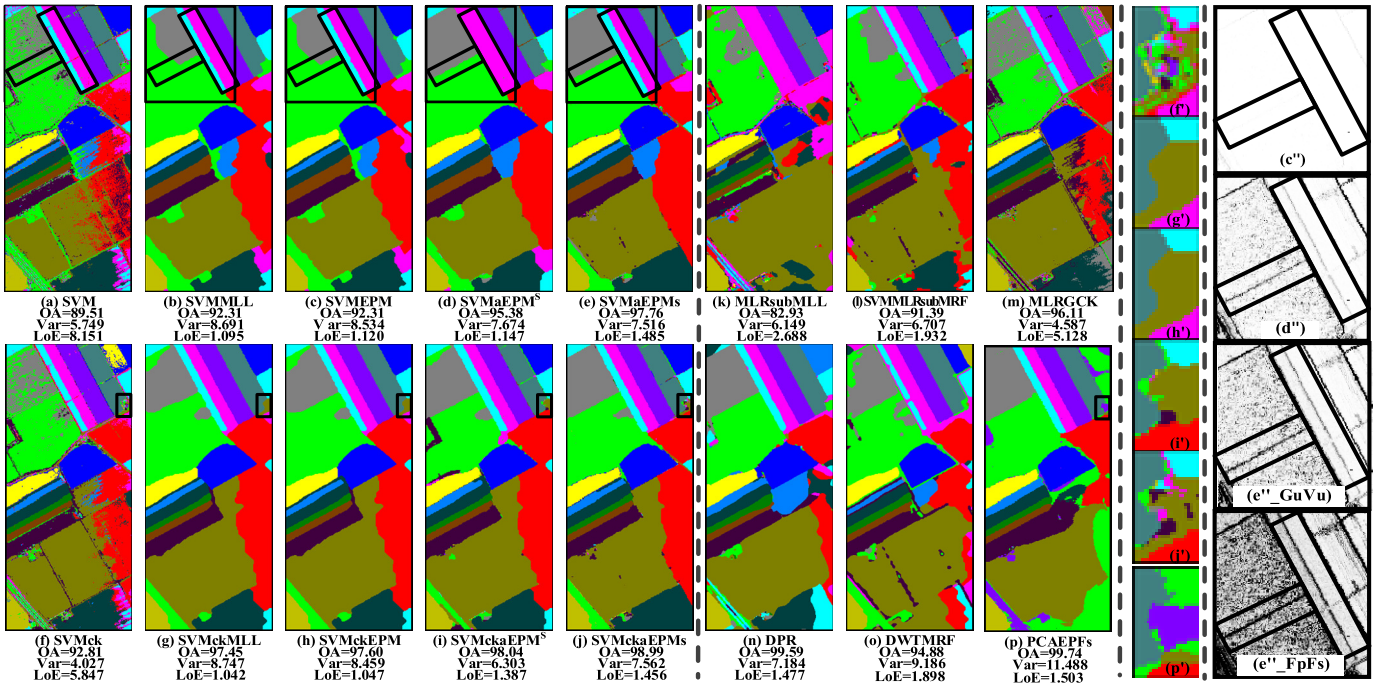


Fig. 4. (a)–(p) Classification maps (OA in %, Var in E+06, and LoE in E+04) on Salinas Scene obtained by different methods. (f′)–(j′), and (p′) Details maps corresponding to (f)–(j) and (p). (c″) EPM for rectangular area in (c). (d″) aEPM for rectangular area in (d). (e″_GuVu) and (e″_FpFs) aEPMs of two class pairs for rectangular area in (e).

1) Fig. 4(a) revealed that noisy pixels appeared in the raw results by SVM. As shown in Table II, compared with SVM, SVMck and MLRGCK improved the OA by 4.23% and 6.58%, respectively. The result maps were smoother and, however, still suffered from noises. Table II shown that SVM, SVMck, and MLRGCK obtained the smallest Var and the largest LoE. This is because the result maps were mixed and over-segmented by misclassified pixels, which instead increased LoE and decreased Var. This suggests the necessity of post-processing on the pixelwise labeling.
2) After MRF correction, MLL (in Table II) promoted OA by 4.14% and 4.29% over SVM and SVMck, respectively. It was observed from Fig. 4(b) and (g) that the refined maps were more consistency, which was visibly in segmentation evaluation metrics. MLL reduced LoE from 7.795 to 1.158 after SVM, and from 6.124 to 1.053 after SVMck. One pixel is so smaller than object, and isolated spots appeared inside class objects. These mislabeled pixels formed false edges and had higher correlation with adjacent segments. This verified the merit of MRF in favoring label consistent. In addition, SVMckMLL improved OA from 93.14% to 97.52% over SVMMLL. SVMck benefited from EMAP, and then promoted the following MRF correction.
3) EPM changed OA slightly compared with MLL, including recall and kappa, as in Table II. These metrics focused on the labeled pixels within class areas instead

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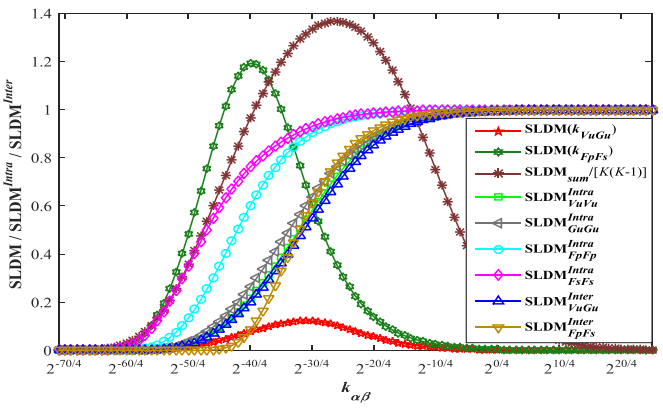


Fig. 5. Effect of $k_{\alpha\beta}$ on SLDM, SLDM^{Intra}, and SLDM^{Inter} for Salinas Scene.

of boundaries and, thus, cannot assess sufficiently MRF refinement. See the segmentation evaluation metric, compared with MLL, EPM reduced Var from 8.379 to 8.230, and increased LoE from 0.911 to 0.918 over SVM, and reduced Var from 8.340 to 8.223, and added LoE from 1.053 to 1.060 after SVMck. Too small μ may favor insufficient smoothness in class areas, and in contrast, too large μ may lead to detail loss. Table I shown that MLL and EPM obtained the best OA when $\mu = 2.00$, which is actually a tradeoff between label consistency and EP. This suggested the need for more efficient aEP schemes in MRF modeling.

4) As observed from Table II, compared with EPM after SVM, aEPM^{S} and aEPMs improved OA from 93.15% to 95.55% and 97.59% and, meanwhile, raised LoE from 1.165 to 1.245 and 1.476, and decreased Var from 8.230 to 7.355 and 7.619, respectively. We may focus on two groups of neighboring class regions, "Vinyard_untrained" (Vu) and "Grapes_untrained" (Gu), and "Fallow_plow" (Fp) and "Fallow_smooth" (Fs). As in Fig. 4(c), EPM improperly regularized pixels of Vu within a connected region as Gu, and as in Fig. 4(d), aEPM^{S} accurately corrected this area as Vu and improved ICA from 57.02% to 77.04%. Fig. 4(d) showed that aEPM^{S}, however, regularized Fp region as Fs falsely and, thus, decreased its ICA from 97.43% to 73.28% (mean over 100 runs). In addition, compared with aEPM^{S}, aEPMs improved ICA of Vu from 77.04% to 89.29%, and ICA of Fp from 73.28% to 97.33%. Overall, aEPMs further increased OA by 2.04% over aEPM^{S}. Fig. 5 showed the change trend of $\mathbf{SLDM}_{\alpha\beta}$ and $\mathbf{SLDM}_{\alpha\alpha}$ by (13) and (14) during one test as $k_{\alpha\beta}$ ranged $[2^{-70/4}, 2^{20/4}]$. Small values of $k_{\alpha\beta}$ make $\mathbf{SLDM}_{\alpha\beta}$ large to detect borders between Vu and Gu, but also make $\mathbf{SLDM}_{\alpha\beta}$ too large, which instead weaken the correction on mislabeled pixels inside Vu or Gu. EPM can be seen as set $k_{\alpha\beta} = 1$, and the edge map by (11) was shown in Fig. 4(c″), where nearly no edge was detected in class areas, but the edges of Vu and Gu were also almost invisible. **SLDM** is defined as the difference between $\mathbf{SLDM}_{\alpha\beta}$ and $\mathbf{SLDM}_{\alpha\alpha}$. For Vu and Gu, aEPM is solved by maximizing **SLDM** for $k_{\alpha\beta} = 2^{-31/4}$, as in Fig. 5. Fig. 4(e″_GuVu) showed that aEPM detected the borders between Vu and Gu clearly and, thus, provided enough edge prior to avoid over correction. Meanwhile, aEPM detected slight false edges in Vu and Gu, so forced enough correction for label uniformity. For the above reasons, aEPMs obtained the highest ICA of Vu after SVM. Moreover, note that aEPM is optimized for each class pair. See Fp and Fs in Fig. 5, **SLDM** is maximized for $k_{\alpha\beta} = 2^{-40/4}$. The related aEPM was shown in Fig. 4(e″_FpFs), as was analogous to Vu and Gu. aEPM^{S} is a tradeoff for all class pairs and Fig. 4(d″) showed the aEPM for $k = 2^{-26/4}$, and the borders of the above classes were much blurrier than that in Fig. 4(e″_GuVu) and (e″_FpFs), respectively. This may lead to over correction on small areas, as in Fig. 4(d).

5) Table II listed that, compared with MLL and EPM after SVMck, aEPM^{S} and aEPMs improved OA from 97.52% to 98.20% and 98.84%, and obvious improvements were seen in the two former class pairs. SVMckaEPMs obtained higher OA than MLRsubMLL, SVMMLRsubMRF. PCAEPFs performed well on Salinas Scene, and obtained the best OA. However, as shown in Table II, SVMck-aEPMs improved LoE from 1.411 to 1.591 compared with PCAEPFs. LoE denotes the goodness of edge preservation. Thus, SVMckaEPMs preserved more edge details than PCAEPFs, as can also be seen from enlarged regions in Fig. 4(j′) and (p′). Var measures the intrasegment homogeneity, and low values of Var mean homogeneous labeling map. Table II listed that PCAEPFs obtained much larger Var than other methods. This is because PCAEPFs resulted in over-smoothness over different class areas and edges, as shown in Fig. 4(p) and (p′). Meanwhile, Table II listed that SVMckaEPMs decreased Var from 11.181 to 7.072 compared with PCAEPFs, which can also be seen in Fig. 4(j) and (j′). Thus, the proposed method has advantages in preserving edge details and favoring area homogeneity. Fig. 5(f′)–(j′) shown the partial enlarged details in Fig. 4(f)–(j). The pixelwise labeling by SVMck in Fig. 4(f′) was mixed with isolated pixels and details. MLL and EPM over refined this area and details lost seriously. aEPM^{S} maintained some edges and aEPMs kept the most details, and Table II listed that aEPMs got the largest LoE compared with other MRF methods.

These above analyses explained the superiority of aEPMs integrating $K(K-1)$ aEPMs in favoring region uniformity and preserving details, including the merit of MRF postprocessing on pixelwise spectral-spatial method, i.e., SVMckaEPMs.

C. Experiments on Pavia University

For Pavia University, a fixed training set (FTS) in Fig. 3(e) is adopted for learning and the remains in Fig. 3(d) excluding these in FTS are used for test. FTS has been utilized in [9], [22], [26], and [28]. The randomly selected training samples over the whole observed scene cannot well model the construction of training set in practical applications [66].

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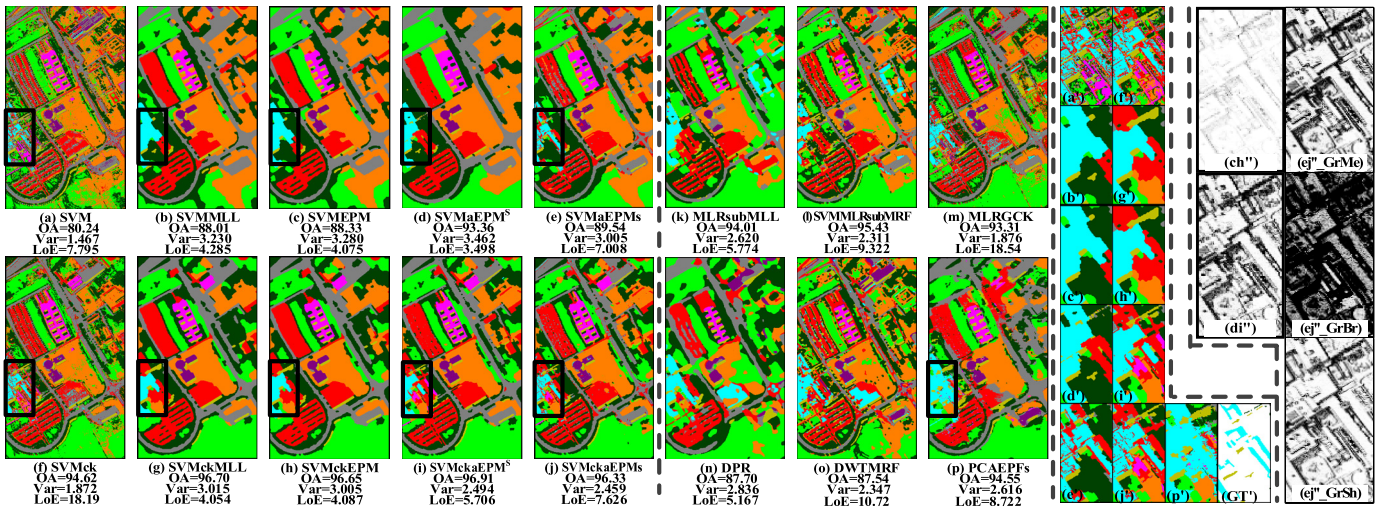


Fig. 6. (a)–(p) Classification maps (OA in %, Var in E+07, and LoE in E+04) on Pavia University obtained by different methods. (a′)–(j′) and (p′) Details maps for (a)–(j) and (p). (ch″) and (di″) EPM for rectangular area in (c) and (h) and (d) and (i), respectively. (ej″_GrMe), (ej″_GrBr), and (ej″_GrSh) aEPMs of three class pairs for rectangular area in (e) and (j).

TABLE III
EVALUATION METRICS ON PAVIA UNIVERSITY OBTAINED BY DIFFERENT METHODS

Class		Train/Test	SVM					SVMck					MLRsub	SVMMLR	MLRGCK	DPR	DWTMRF	PCA
			Raw	MLL	EPM	aEPM^S	aEPMs	Raw	MLL	EPM	aEPM^S	aEPMs	MLL	subMRF				EPFs
C1	Asphalt	548/6304	83.85	98.37	98.37	**100**	93.45	93.56	**99.89**	**99.89**	99.60	95.32	93.05	94.33	94.99	81.35	79.77	81.28
C2	**Meadows**	540/18146	68.94	77.82	76.90	**91.19**	84.92	97.41	99.66	99.69	**99.70**	99.64	98.97	99.44	98.41	94.61	85.80	97.69
C3	**Gravel**	392/1815	68.98	**80.06**	77.58	66.17	59.12	**54.10**	39.83	38.95	39.89	43.97	66.34	42.37	54.10	81.43	76.53	**89.09**
C4	Trees	524/2912	**98.52**	96.60	98.25	91.00	96.77	98.97	98.52	98.52	**99.69**	99.62	80.05	97.28	95.91	91.55	98.97	94.47
C5	Metal sheets	265/1113	99.37	**100**	**100**	99.91	99.91	99.28	**100**	**100**	99.91	99.91	**100**	99.93	99.82	95.96	96.59	99.55
C6	Bare Soil	532/4572	93.83	**100**	**100**	**100**	**100**	94.12	**100**	**100**	**100**	99.98	98.91	98.67	79.90	61.57	88.23	**100**
C7	Bitumen	375/981	90.32	98.88	98.88	**99.18**	98.06	97.76	**100**	**100**	**100**	98.47	54.13	68.57	95.82	79.41	97.45	92.25
C8	**Bricks**	514/3364	92.72	98.25	98.25	**98.54**	97.92	97.56	98.31	98.31	**99.97**	99.82	97.68	99.47	96.11	96.02	98.69	96.64
C9	**Shadows**	231/795	**98.36**	82.39	91.32	84.40	86.16	**96.10**	92.20	91.57	92.96	92.96	99.87	**99.89**	96.73	94.34	95.85	96.35
OA (Mean in %)			80.24	88.01	88.33	**93.36**	89.54	94.62	96.70	96.65	**96.91**	96.33	94.01	95.43	93.31	87.70	87.54	94.55
Recall/AA (Mean in %)			88.32	92.48	**92.86**	92.27	90.70	92.10	92.05	91.88	**92.41**	92.19	87.67	88.88	90.20	86.25	90.88	**94.15**
Kappa			75.00	84.53	84.95	**91.18**	86.29	92.76	95.53	95.47	**95.81**	95.05	91.85	93.94	90.90	83.23	83.65	92.68
Var (Mean in E+07)			1.467	3.230	3.280	3.462	**3.005**	1.872	3.015	3.005	2.494	**2.459**	2.620	2.311	1.876	2.836	2.347	2.616
LoE (Mean in E+04)			25.31	4.285	4.075	3.498	**7.008**	18.19	4.054	4.087	5.706	**7.626**	5.774	9.322	18.537	5.167	10.719	8.722

And this strategy may reduce the classification difficulty and cannot well evaluate the generalization ability of the classifier [66]. Thus, this test set is more in line with the real training process. The parameter related to SVM, SVMck, EMAP, k^{opt}, and $k^{\text{opt}}_{\alpha\beta}$ is selected as these for Salinas Scene. Table I listed the values μ tuned for compared MRF methods for the best OA.

Fig. 6 and Table III present the labeling maps and evaluation metrics. Detailed experimental analysis is as follows.

1) Table III showed that with EMAP, SVMck and MLR-GCK increased OA by 14.38% and 13.77% compared with SVM. The result maps by SVMck and MLRGCK still suffered from isolated misclassified pixels. Fig. 6 revealed that the MRF-based schemes obtained smoother labeling within class areas than pixelwise methods. This verified again the merit of MRF smoothness on pixelwise labeling.
2) As shown in Table III, after SVM-based labeling, aEPM^S and aEPMs promoted OA by 5.03% and 1.21% compared with EPM. In detail, aEPM^S promoted ICA of "Asphalt" from 98.37% to 100% and ICA of "Bitumen" from 98.88% to 99.18% over MLL and EPM. aEPMs promoted ICA of "Meadows" (Me) from 76.90% to 84.92% over EPM. These improvements in accuracy showed again the superiority of the proposed methods.
3) In accuracy metrics, however, the increase of some classes, e.g., "Asphalt" and "Bitumen" showed accuracy saturation of the compared methods with limited test samples. First, the test set has a small proportion over the size of remote sensing scene. Second, these limited pixels are inside class areas other than near borders. Thus, the visual observation on labeling maps is needed, and Fig. 6(a′)–(e′) showed the enlarged details of label maps in Fig. 6(a)–(e) for SVM, and further refined by MLL, EPM, aEPM^S, and aEPMs. This region contains "Gravel" (Gr), Me, "Bricks" (Br), "Shadows" (Sh), and so on. Note that after SVM, MLL obtained the highest ICA of Gr, as illustrated in Table III. This increase, however, is at the cost of over-regularizing large areas in Fig. 6(b′) as Gr and serious detail lost, e.g., the land-cover Sh. See that SVM obtained the best ICA of Sh, 98.36%, in Table III, and identified lots of Sh details in Fig. 6(a′). However, Sh lies where light is blocked by trees and buildings, which is often surrounded by other classes and has small sizes, as shown in Fig. 3(c). Thus, Sh has high risk to be over-refined in

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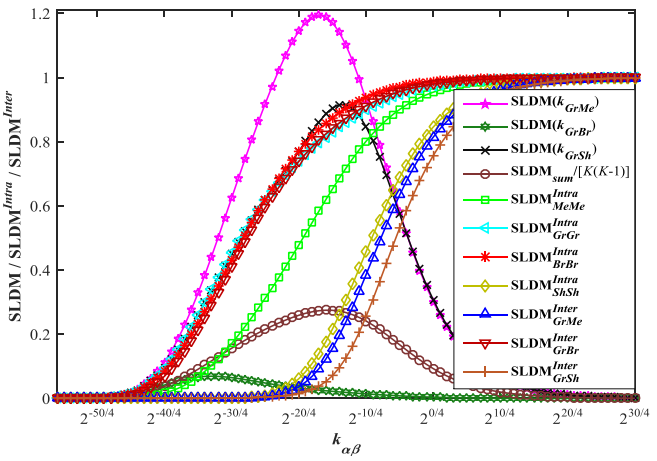


Fig. 7. Effect of $k_{\alpha\beta}$ on SLDM, SLDM^{Intra}, and SLDM^{Inter} for Pavia University.

MLL correction, as shown in Fig. 6(b′). Table III showed that MLL decreased the ICA of Sh from 98.36% to 82.39% after SVM, in which case the issue is to preserve the Sh details. In Fig. 6(c′), EPM, still improperly corrected lots of Sh details as Gr, for the EPM by (10) in Fig. 6(ch″) hardly provided edge prior. In contrast, aEPM^{S} and aEPMs protected Sh details better than EPM and, meanwhile, corrected the isolated mislabeled pixels after SVM, as shown in Fig. 6(d′) and (e′). The edge map by (16) was shown in Fig. 6(di″) for $k^{\text{opt}} = 2^{-15/4}$, and the edge map by (17) for Gr and Sh was shown in Fig. 6(ej″_GrSh) for $k_{\alpha\beta}^{\text{opt}} = 2^{-13/4}$. The two edge maps solved by SLDM considers comprehensively the interclass/intraclass label dependence of nearby pixels and achieved superior tradeoff between label uniformity and detail preservation.

4) As observed in Fig. 6(d′) and (e′), aEPMs protected more details of Me and Br that aEPM^{S}, whereas they preserved nearly the same details of Sh. Fig. 7 illustrated the trend of **SLDM**, **SLDM**$_{\alpha\beta}$, and **SLDM**$_{\alpha\alpha}$ by (13)–(15) when $k_{\alpha\beta}$ ranged $[2^{-50/4}, 2^{30/4}]$. aEPM^{S} is a simplification of aEPMs and is a tradeoff for all possible adjacent classes. Here, the edge map by (16) when $k^{\text{opt}} = 2^{-15/4}$ happened to be similar to that by (17) for Gr and Sh when $k_{\alpha\beta}^{\text{opt}} = 2^{-13/4}$, as shown in Fig. 6(di″) and (ej″_GrSh). Thus, the preservation effects on Sh by aEPM^{S} and aEPMs were similar. This edge map may not be the optimal selection for other classes, e.g., Me and Sh, which were over-smoothed seriously in Fig. 6(d′). In Fig. 6(ej″_GrMe-ej″_GrBr), the class-by-class adaptive edge maps provided more edge information in preserving details of Me and Br in Fig. 6(e′). This verified the merit of aEPMs in combining $K(K-1)/2$ aEPMs in EP schemes.

5) After SVMck, the OA improvement by compared MRF schemes was nearly the same. As in Table III, OAs by aEPM^{S} and aEPMs after SVMck were obviously higher than these by DWTMRF. Table III listed that compared with PCAEPFs, SVMckaEPM^{S} and SVMck-aEPMs improved the OA accuracy from 94.55% to 96.91% and 96.33%, respectively. Moreover, SVMck-aEPMs preserved more edges and details, as shown in Fig. 6(j′) and (p′). The use of segment evaluation metrics covered again the shortages of classical evaluate accuracy with limited test pixels. As shown in Table III, SVMck-aEPM^{S} and SVMckaEPMs decreased Var from 2.616 to 2.494 and 2.459, respectively. This is due to the proposed methods favored more homogenous labeling results and avoided over-smoothness over different classes. This verified again the superiority of aEPMs in preserving edges and favoring region uniformity.

D. Experiments on DFC2018sub

For DFC2018sub, 50 samples were randomly selected from each class for training, and the remains were for test. For simplicity, the parameters of SVM, SVMck, and EMAP were selected or optimized as these for Pavia University. k^{opt} and $k_{\alpha\beta}^{\text{opt}}$ were solved using the SLDM measure. Table I showed the values μ for the compared MRF methods with the best OA.

The labeling maps and evaluation metrics are shown in Fig. 8 and Table IV. Detailed experimental analysis is as follows.

1) Table IV showed that with EMAP, SVMck and MLRGCK increased OA by about 10% and 9% compared with SVM. It can be seen from Fig. 8 that the result maps by SVMck and MLRGCK suffered from isolated mislabeled pixels. As in Fig. 8, the MRF-based schemes shown smoother labeling within class regions than pixelwise approaches. This presents the effect of MRF smoothness on pixelwise labeling.
2) After MRF regularization, MLL (in Table IV) promoted OA about 16% and 9% over SVM and SVMck. It was seen from Fig. 8(b) and (g) that the corrected maps were smoother, as was visibly in LoE. MLL reduced LoE from 39.747 to 5.880 after SVM, and from 34.425 to 7.284 after SVMck. Isolated misclassified pixels appeared inside class objects. These mislabeled samples formed false edges and had higher correlation with adjacent segments. This shown the merit of MRF in favoring region smoothness. In addition, SVMckMLL improved OA from 78.03% to 81.90% over SVMMLL. SVMck benefited from EMAP, and then promoted the following MRF smoothness.
3) EPM changed OA slightly compared with MLL after SVM and SVMck, including recall, and kappa, as shown in Table IV. These metrics cannot assess sufficiently the effect of MRF for they focused on the labeled pixels within class regions instead of boundaries. See the segmentation evaluation metric, compared with MLL, EPM reduced Var from 8.379 to 8.230, and increased LoE from 0.911 to 0.918 over SVM, and reduced Var from 8.340 to 8.223, and added LoE from 11.102 to 11.064 after SVM. Too large μ may lead to detail loss, while too small μ may not favor enough correction within class areas. In contrast, Table IV also shown

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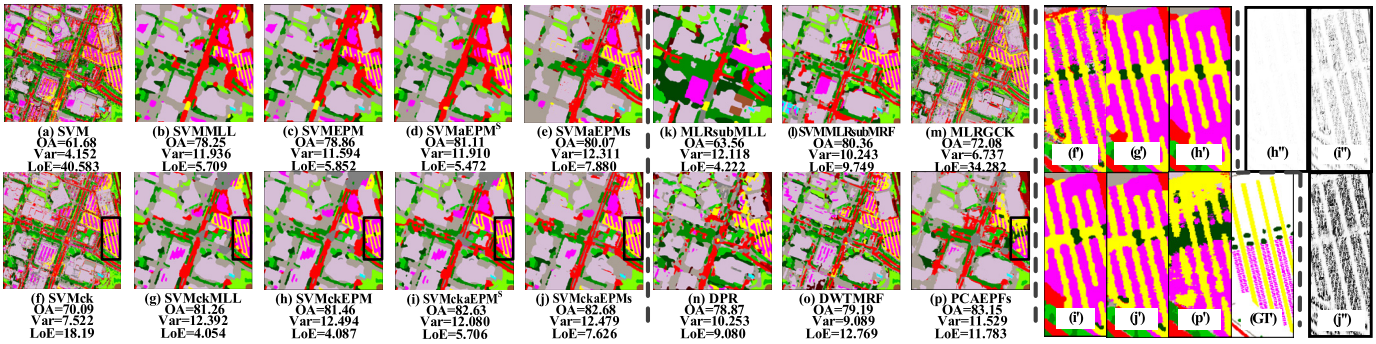


Fig. 8. (a)–(p) Classification maps (OA in %, Var in E+07, and LoE in E+04) on DFC2018sub obtained by different methods. (f′)–(j′) and (p′) Details maps corresponding to (f)–(j) and (p). (h″) and (i″) EPM for rectangular area in (h) and (i). (j″) aEPM of class pairs, "Paved parking lots" and "Cars," for rectangular area in (j).

TABLE IV
EVALUATION METRICS ON DFC2018SUB OBTAINED BY DIFFERENT METHODS

Class		Train/Test	SVM					SVMck					MLRsub	SVMMLR	MLRGCK	DPR	DWTMRF	PCA
			Raw	MLL	EPM	aEPM^S	aEPMs	Raw	MLL	EPM	aEPM^S	aEPMs	MLL	subMRF				EPFs
C1	Healthy grass	50/2710	**95.16**	81.30	81.34	79.09	63.38	**94.46**	83.13	82.39	81.77	74.67	53.04	77.67	92.33	85.29	87.56	85.39
C2	Stressed grass	50/3641	92.45	91.57	91.77	**92.72**	86.09	92.06	91.78	91.69	**93.06**	91.68	80.31	92.05	90.09	85.33	91.49	83.44
C3	Evergreen trees	50/4546	90.89	94.24	94.37	**94.86**	86.16	94.60	96.50	96.17	**97.30**	93.17	96.40	**98.75**	94.66	95.29	96.99	95.68
C4	Deciduous trees	50/1206	**90.87**	73.54	73.80	70.41	58.64	**93.17**	79.29	75.97	81.71	70.24	64.32	90.44	**93.59**	87.83	94.31	97.52
C5	Bare earth	50/271	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
C6	Water	50/97	**99.82**	99.31	99.33	99.67	98.49	**100**	99.88	99.82	**100**	99.61	99.52	**100**	99.77	99.92	99.58	99.93
C7	Non-residential buildings	50/110179	61.62	82.94	83.07	84.82	**84.89**	73.21	86.79	87.13	**87.50**	**87.83**	69.48	83.71	71.15	82.51	79.86	**89.58**
C8	Roads	50/14373	50.11	49.55	50.05	**55.00**	54.42	**64.72**	58.31	57.37	63.68	64.71	17.87	54.69	**66.60**	54.21	63.14	59.45
C9	Sidewalks	50/13937	45.36	**56.53**	56.10	52.42	55.87	48.20	**56.40**	56.17	52.51	51.06	29.32	53.02	52.43	39.29	**68.39**	50.22
C10	Crosswalks	50/827	**61.71**	8.41	8.35	4.85	5.60	**67.73**	18.56	15.92	16.23	16.88	33.49	40.45	71.20	75.21	59.90	**86.39**
C11	Major thoroughfares	50/1762	65.88	**94.79**	94.45	87.73	88.41	89.62	**95.48**	95.29	88.43	87.29	97.53	94.12	92.50	96.30	93.95	**97.11**
C12	Highways	50/700	**81.03**	73.14	75.04	76.09	66.67	98.61	99.06	98.94	**99.17**	98.21	16.17	84.77	98.38	96.97	94.03	**99.64**
C13	Paved parking lots	50/3424	92.57	87.94	88.01	73.90	**93.18**	**96.14**	92.37	90.36	91.43	95.55	10.85	96.66	96.96	88.48	97.36	90.33
C14	Cars	50/1957	81.82	**93.74**	93.70	91.51	85.65	87.67	**95.47**	95.13	94.37	88.87	96.33	94.18	86.08	87.34	95.91	86.07
OA (Mean in %)			62.62	78.03	78.14	**79.03**	78.88	72.81	81.90	81.92	**82.46**	82.30	61.05	79.32	71.90	77.11	79.08	**83.56**
Recall/AA (Mean in %)			**79.23**	77.64	77.81	75.93	73.39	**85.73**	82.36	81.60	81.94	79.98	61.76	82.89	86.12	83.86	87.32	**87.20**
Kappa			45.08	62.09	62.25	**63.21**	62.92	56.49	67.76	67.68	**68.56**	68.13	40.53	64.37	55.59	60.82	64.88	**69.90**
Var (Mean in E+07)			4.912	11.102	**11.064**	11.403	11.656	7.284	11.464	11.671	**11.355**	11.555	11.454	9.829	6.164	9.541	10.152	10.903
LoE (Mean in E+04)			39.747	5.880	5.950	5.768	**7.694**	34.425	6.444	6.009	6.902	**8.222**	4.115	9.527	35.98	9.123	13.134	12.696

that MLL and EPM obtained the best OA after SVM when $\mu = 2.00$. This is a tradeoff between EP and label consistency. This suggested the need for more efficient aEP schemes in MRF modeling.

4) As in Table IV, aEPM^S improved the recall of class "Roads" from 50.05% to 55.00%, and the recall of class "Stressed grass" from 91.77% to 92.72% compared with EPM after SVM. After SVMck, aEPM^S improved the recall of class "Stressed grass" from 91.69% to 93.06% compared with EPM. The improvement in OA by aEPM^S and aEPMs compared with EPM may be not obvious. aEPM^S improved OA by about 0.9% compared with EPM after SVM and by nearly 0.5% after SVMck. This is because the parameter μ for different MRF methods was tuned when they obtained the highest OA. As in Table I, μ in aEPMs was tuned as 7.25 and 4.75 for SVM and SVMck. aEPM^S and aEPMs obtained a tradeoff between favoring label smoothness and edge-preservation. The value of μ may be too large for the classes of small areas, and over correction still happens within these land covers.
5) Both aEPM^S and aEPMs after SVMck performed better than DWTMRF. PCAEPFs performed well for labeling "Nonresidential buildings" areas, which were large class regions and had over 69% samples of the test set. However, Table IV showed that SVMckaEPMs increased the recall of "Paved parking lots" from 90.33% to 95.55% compared with PCAEPFs. aEPMs improved the recall of class "Paved parking lots" from 88.01% to 93.18% compared with EPM after SVM and from 90.36% to 95.55% after SVMck. As in Fig. 8(e′), (j′), and (p′), the proposed methods preserved the "Paved parking lots" regions better than the other MRF methods and PCAEPFs and, meanwhile, promoted more homogeneous labeling maps. Meanwhile, aEPMs got the highest LoE after SVM and SVMck, when for aEPMs, the largest μ was tuned. Consequently, aEPM^S and aEPMs have superiority in preserving edges and region uniformity, especially for edges in Salinas Scene, small areas in Pavia University, and "Paved parking lots" in DFC2018sub.

E. Parameter Analysis and Running Time

This section analyzes the effect of μ and λ on the compared Markovian methods, after SVM and SVMck. Figs. 9–12 showed the changes of OA, Var, and LoE, as μ and λ increased. The other settings were the same as that in Section IV.

1) As shown in Figs. 9–11(a) and (b). OAs by MLL, EPM, aEPM^S, and aEPMs improved gradually as μ started to increase at first, and then fell back. In MRF

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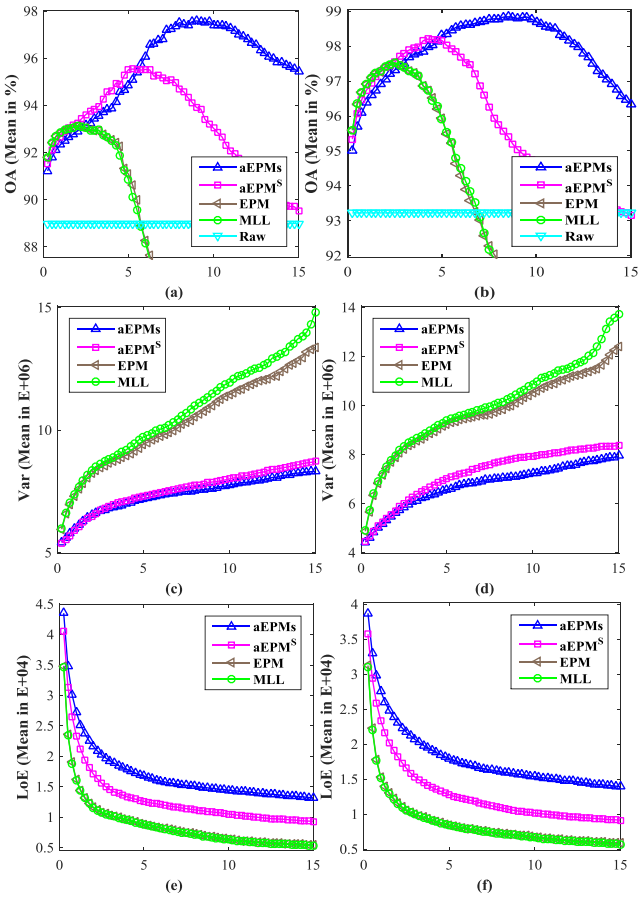


Fig. 9. Effect of μ on MRF after SVM, and SVMck for Salinas Scene. (a), (c), and (e) μ in MRF for SVM on Salinas Scene. (b), (d), and (f) μ in MRF for SVMck on Salinas Scene.

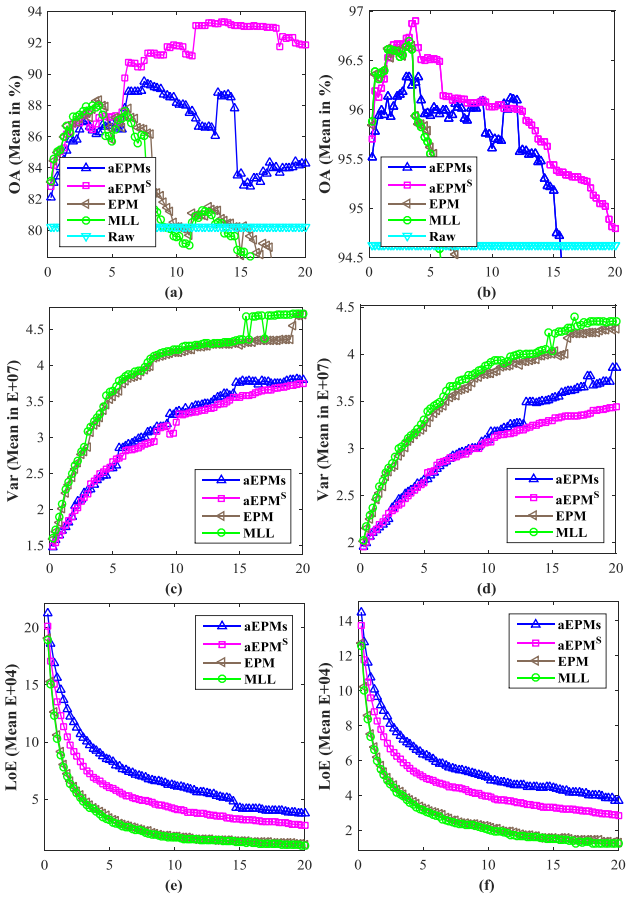


Fig. 10. Effect of μ on MRF after SVM, and SVMck for Pavia University. (a), (c), and (e) μ in MRF for SVM on Pavia University. (b), (d), and (f) μ in MRF for SVMck on Pavia University.

regularization, μ tunes the mutual effect of the data and smoothness terms. With an initial increase in μ, the smoothing effect just began to work and the mislabeled pixels were refined. The best OAs arisen when μ was just proper to correct most mislabeled pixels and to avoid serious over-correction of small regions. However, as μ continued to increase, the over-smoothing risk raised and, therefore, OAs fell back.

2) See the OAs illustrated in Fig. 9(a), aEPMs, aEPM^S, EPM, and MLL got the best OAs, 97.59%, 95.55%, 93.15%, and 93.14%, respectively, when μ was 9.00, 5.25, 2.00, and 2.00. The μ value when aEPMs and aEPM^S obtained the best OA was larger than that of EPM and MLL. The value of μ tunes the strength of smoothness prior, and too large values of μ increase the risk of over-smoothing and detail loss. EPM and MLL, however, provided few edges prior and cannot resist fully the over-smoothing risk and, hence, obtained the best OA with relatively small μ. In contrast, aEPMs and aEPM^S used the aEPM optimized by SLDM and got a better tradeoff between favoring label uniformity and preserving details with relatively large μ.

3) The classical evaluate metrics are confined by the limited available test pixels inside homogenous land covers. The segmentation quantity metrics, Var and LoE, covered this shortage, of which the changing curves by different MRF methods were shown in Figs. 9–11(c)–(f) when μ increased. First observe the common trend of changing curves. See that the raw pixelwise results got the smallest Var and the largest LoE. The pixelwise labeling maps were over-segmented severely by misclassified isolated pixels and, hence, were most intrasegment homogeneous and kept many false edges inside land covers. After an initial increase in μ, the smoothness prior began to take effect and refined the mislabeled spots. The refined maps were not as intrasegment homogeneous as the pixelwise mapping. This leaded to increase in Var in Figs. 9–11(c) and (d). Besides, this removed the false edges around the mislabeled pixels and obvious declines were observed in LoE in Figs. 9–11(e) and (f). As μ continued to add, the effect of MRF smoothness went through three states, under-smoothing, proper correction, and over-refinement. The over-smoothing status erroneously modified small details and edges, and increased Var and decreased LoE.

4) Then observe the difference among changing curves by compared MRF methods. As shown in Figs. 9–11, for the same values of μ, aEPMs and aEPM^S obtained lower Var and larger LoE than EPM and MLL. Small Var means good intrasegment homogeneity and

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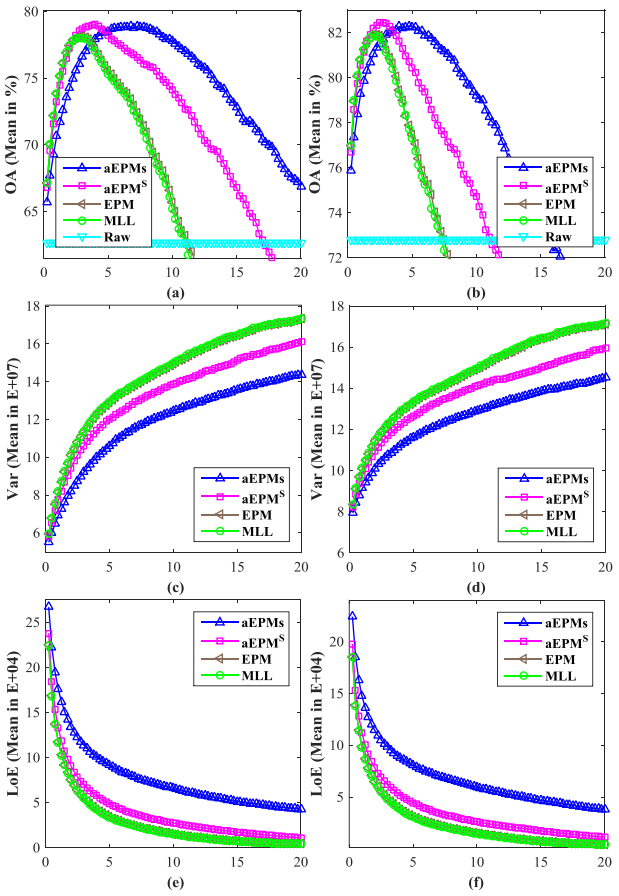


Fig. 11. Effect of μ on MRF after SVM, and SVMck for DFC2018sub. (a), (c), and (e) μ in MRF for SVM on DFC2018sub. (b), (d), and (f) μ in MRF for SVMck on DFC2018sub.

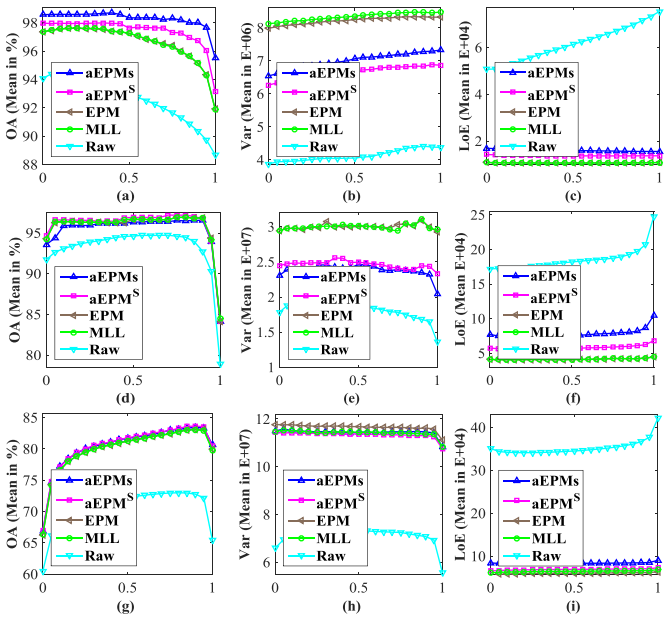


Fig. 12. Effect of λ on MRF after SVMck on three HSI data. (a)–(c) λ in SVMck on Salinas Scene. (d)–(f) λ in SVMck on Pavia University. (g)–(i) λ in SVMck on DFC2018sub.

large LoE indicates effective preservation of details and edges. This verified again the superiority of aEPMs in favoring region label consistency and preserving details.

TABLE V
RUNNING TIME OF COMPARATIVE METHODS

Running Time (s)		Salinas Scene	Pavia University	DFC2018sub
SVM	Raw	12.47	38.73	22.56
	MLL	15.80=12.47+ 3.33	42.96=38.73+ 4.23	41.65=22.56+ 19.09
	EPM	23.35=12.47+10.88	56.21=38.73+17.48	61.73=22.56+ 39.17
	aEPM^S	15.96=12.47+ 3.49	68.87=38.73+30.14	39.07=22.56+ 16.51
	aEPMs	110.32=12.47+97.85	120.14=38.73+81.41	255.07=22.56+232.51
SVMck	Raw	136.16	1883.95	105.28
	MLL	139.33=136.16+ 3.17	1888.57=1883.95+ 4.62	119.75=105.28+ 14.47
	EPM	147.15=136.16+10.99	1901.96=1883.95+18.01	141.09=105.28+ 35.81
	aEPM^S	139.61=136.16+ 3.45	1914.81=1883.95+30.86	120.81=105.28+ 15.53
	aEPMs	232.67=136.16+96.51	1965.20=1883.95+81.25	347.09=105.28+241.81
MLRsubMLL		8.86	11.98	28.88
SVMMLRsubMRF		29.72	58.69	12.35
MLRGCK		5.14	176.36	2.32
DPR		147.86	169.03	408.56
DWTMRF		1470.14	2247.89	3263.86
PCAEPFs		14.23	132.45	35.263

5) The parameter λ tunes the spectral and EMAP features in SVMck and thus affects the raw pixelwise labeling. The role of spectral feature increased when λ changed from 0 to 1, while the role of EMAP decreased. As in Fig. 12(a), (d), and (g), the OAs by SVMck generally increased at first and then fell back, which reveals the superiority of fusing spectral-spatial features in HSI classification.
6) Fig. 12(a), (d), and (g) shown that all MRF methods increased the OAs after SVMck. The increase was obvious (6%–9%) on DFC2018sub for different λ values. The OAs of the MRF methods were more robust to different λ values than that of SVMck, especially for Pavia University, and Salinas Scene. This revealed the advantages of combining MRF and pixelwise spectral–spectral labeling approaches.
7) As in Fig. 12(b), (e), and (h), the Vars of MRF methods had the same changing trends as that of SVMck. Var measures the intrasegment homogeneity. The homogeneity of MRF-refined result was affected by the raw SVMck labeling. aEPM^S and aEPMs got smaller Var than MLL and EPM, thus the proposed methods performed better in favoring segment homogeneity and avoid over-correction.
8) The LoE by SVMck increased for more spectral content was fed to SVMck when λ ranged [0,1]. EMAP is useful to force smoother labeling, and the decrease of EMAP leaded to more misclassified pixels and more false edges inside class regions, and thus, the LoE increased when λ ranged [0,1]. Meanwhile, aEPM^S and aEPMs got larger LoE than MLL and EPM, which means that the proposed methods have superiority in preserving edges and small details.

The energy function of aEPMs and aEPM^S is solved by the $\alpha\beta$-swap algorithm, which was proved to terminate in $O(n)$ [53], and reported completion of improvements after a few cycles [51]. In the experiments, the cycle number was set as $N^{\text{total}} = 5$. The comparative approaches were conducted on a computer with 3.30-GHz Inter i5-4590 and 8.00-GB RAM, and the running time (mean values over ten random runs) of compared methods was shown in Table V. Though it took more time to obtain the raw result, the integration of aEPMs and aEPM^S with SVMck improved the classification results than that after SVM.

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V. Conclusion

This article presents a novel aEPMs scheme for MRF-based HSI classification. For each class pair, the corresponding aEPM is optimized by the designed SLDM criterion. aEPMs aims to achieve a good compromise between region uniformity and detail preservation. Experimental results conducted on HSIs and the analysis by segmentation metrics reveals the superiority of aEPMs in classification accuracy, favoring label smoothness, and preserving edges. The future work is as follows.

1) The integration of aEPMs with SVMck further improved the labeling result than that after SVM. The integration of aEPMs and other efficient methods, i.e., MLRGCK [64], multiple feature learning (MFL) [69], is expected.
2) In the presented SLDM, SAM is computed on the original spectral feature to describe SSM. Inspired by [37] in which SID is computed on the 3-D DWT-based feature. Using more efficient SSM on extracted spectral–spatial feature is expected to increase interclass variation and reduce intraclass variability.
3) Integrating aEPMs with TV regularization [38] and other adaptive-MRF schemes (multiscale [39], multiresolution [40], and hierarchical [42]) is our future work. Meanwhile, MRF classification belongs to postprocessing methods. Similarly, edge detection and EP are also one key issue in other postprocessing methods [21], [22]. Our future work may include the integration of aEPMs with object-oriented methods and relearning framework [70], [71].
4) The ideal of aEPMs may be extended and integrated with superpixel [72], hierarchical [73], and multiscale [74] deep neural network-based framework.

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